

Coordination complex (HL)

Higher level (HL)

Learning outcomes

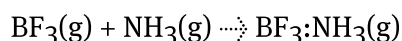
At the end of this section, you should be able to:

- (R3.4.7) Draw and interpret Lewis formulas of reactants and products to show coordination bond formation in Lewis acid—base reactions.
- (R3.4.8) Deduce the charge on a complex ion, given the formula of the ion and ligands present.

In [section S2.2.1a](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43106/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43106/>), a Lewis formula is used to show all valence electrons involved representing the covalent bonding. This formula also can be used to show the bond that is formed due to the shared pair of electrons from the same atom, known as a coordination bond (see [section S2.2.3](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/coordination-bonds-id-43594/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/coordination-bonds-id-43594/>)). This bond can also be formed between a metal ion and ligand by sharing of a pair of electrons, which often forms brightly coloured compounds. For instance, $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ forms a bright blue compound. The concept of sharing a pair of electrons is also connected with the previous [section R3.4.1–2](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/nucleophilic-substitution-reaction-id-43490/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/nucleophilic-substitution-reaction-id-43490/>), in which nucleophiles donate a pair of electrons, which is also called Lewis base (see later in this section). What is the relationship between a coordination bond with a nucleophile and Lewis base? What is their connection?

Lewis acid—base interaction: formation of coordination bond

The interaction between a Lewis acid and Lewis base results in a formation of a coordination bond as an electron pair that is donated from the Lewis base coming from the same atom. This typical bond is always produced in a Lewis acid–base reaction. For example, ammonia donates a lone pair as Lewis base to boron trifluoride as Lewis acid because it has an incomplete octet (**Figure 1**).



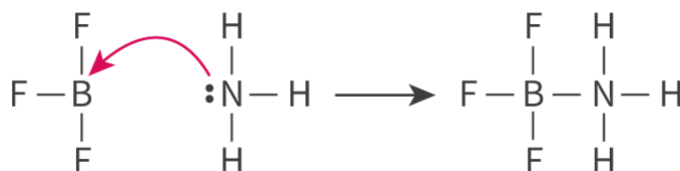


Figure 1. A lone pair from ammonia is donated to boron trifluoride.

 More information for figure 1

The image is a chemical reaction diagram illustrating the interaction between ammonia (NH₃) and boron trifluoride (BF₃). On the left side, the structure of BF₃ is shown with boron (B) at the center and three fluorine (F) atoms bonded to it. Ammonia (NH₃) is depicted with nitrogen (N) at the center, bonded to three hydrogen (H) atoms. A lone pair of electrons is represented as a pair of dots adjacent to the nitrogen. An arrow points from the lone pair to the boron atom, indicating the donation of electrons to form a coordinate bond. On the right side, the resulting compound is displayed as F-B-N-H, with B now having a coordinate bond to N, signifying the formation of a new adduct.

[Generated by AI]

The nitrogen atom (in ammonia) donates a pair of the non-bonding electrons to a boron atom that has an incomplete valence shell. What does this process have in common with nucleophilic attack?

A nucleophile is attracted to an electrophile and donates an electron pair forming a new covalent bond (see [section R3.4.1–2](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/nucleophilic-substitution-reaction-id-43490/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/nucleophilic-substitution-reaction-id-43490/>)). This process is similar to the Lewis acid–base interaction in which the formation of a new bond due to an electron pair is donated by the Lewis base. However, the bond that is formed due to this interaction is called a coordination bond.

Making connections

The properties of coordination bonds compared to other covalent bonds are similar.

Once the coordination bond is formed, it is indistinguishable from other covalent bonds. The only difference is the way in which the bond is formed. The coordination bonds are formed by sharing a pair of electrons from the same atom (see [section R3.2.1–2](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/coordination-complex-hl-id-46330/print/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/coordination-complex-hl-id-46330/print/>)).

[753301/book/oxidation-and-reduction-id-43487/](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/oxidation-and-reduction-id-43487/)), whereas the bonds of other covalent bonds are formed by electrons that are shared equally from the two atoms (see [section S2.2.1a](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43106/>)).

Therefore, this is similar to the process that occurs between the nucleophilic attack and Lewis acid–base interaction involving donation and acceptance of an electron pair. In other words, a Lewis base is also called a nucleophile and a Lewis acid is an electrophile. Another example of the formation of coordination bonds is illustrated in **Figure 2** in which an electron pair from nitrogen is donated to hydrogen ions.

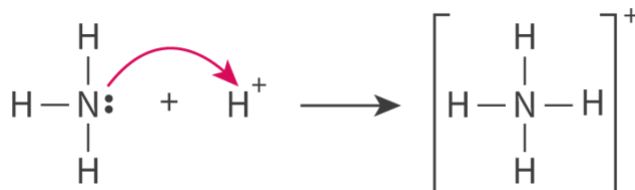


Figure 2. Ammonia protonation.

 More information for figure 2

The image is a chemical diagram illustrating ammonia protonation. It depicts a structural formula where nitrogen (N) is bonded to three hydrogen (H) atoms, presented with a lone pair represented by a pair of dots next to nitrogen. An arrow points from the lone pair to a hydrogen ion (H⁺), indicating the donation of the electron pair from nitrogen to the hydrogen ion. This forms a new bond, shown on the right part of the diagram, where the nitrogen is now positively charged and forms a coordinate bond with the additional hydrogen, resulting in an ammonium ion (NH₄⁺). The ammonium ion is enclosed in square brackets with a positive charge indicating its formation.

[Generated by AI]

The charge on a complex ion

In [section 3.1.8](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-2-id-43595/>), transition elements are described as having incomplete d-sublevels that give them characteristic properties such as the formation of complex ions with ligands. For example, $[\text{Fe}(\text{CN})_6]^{4-}$ naturally forms bonds between ligands and the transition element. How does this relate with the Lewis acid–base interaction? Which is acting as the Lewis acid and which is acting as the Lewis base?

Iron has unoccupied d-orbitals accepting the non-bonding pair of electrons from the cyanide ion, CN^- . As it involves acceptance and donation of an electron pair, the formation of ion complex is related to

Lewis acid—base interaction. Iron is identified as a Lewis acid and cyanide ion is identified as a Lewis base.

Another example is the formation of $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$ complex ions. Copper(II) ions (Cu^{2+}) in aqueous solution receive an electron pair from water as a Lewis base or ligand (**Figure 3**). This is because copper(II) is surrounded by water molecules as ligands.

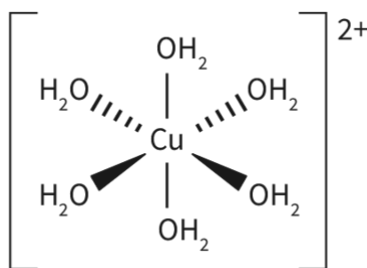
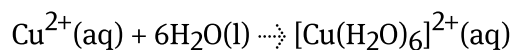


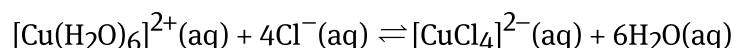
Figure 3. The $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$ complex ion.

 More information for figure 3

This is a structural diagram of the copper hexaaqua complex ion, $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$. At the center of the diagram is a copper ion (Cu). Surrounding the central Cu are six water molecules, each denoted as H_2O . These water molecules are arranged in an octahedral geometry around the copper ion. The bonds between the copper ion and the water molecules are depicted with different lines: solid lines for in-plane bonds, dashed lines for bonds extending outwards, and bold lines for bonds going inwards. This geometric arrangement illustrates the spatial orientation of the water molecules relative to the central copper ion, showcasing the characteristic structure of the complex ion.

[Generated by AI]

Complex ions may experience a substitution reaction such as water in aqueous solution being replaced by other ligands.



The number of coordination bonds formed by central metal ions and ligands is called the coordination number. The strength and stability of the coordination bond in complex ions is affected by the net charge of both ligands and the central metal ion. The electrostatic force will be strong in complex ions if both the opposite charges from ligands and the central metal ion are high. Neutral ligands such as H_2O , NH_3 and CO have a charge of zero, and negatively charged ligands such as Cl^{-} ,

CN^- and Br^- have a charge of minus one (-1). To help you understand how to deduce the charge on a complex ion, **Figure 4** shows the components of one in detail.

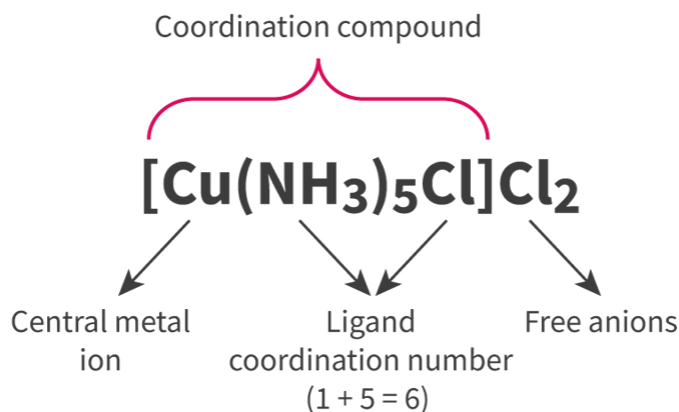


Figure 4. The formula of ions and ligands in a complex ion.

 More information for figure 4

This diagram illustrates the formula of a complex ion, specifically focusing on the coordination compound. At the center of the diagram is the formula $[\text{Cu}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$, which represents a copper complex ion. Above the formula, there is a label "Coordination compound" with a bracket indicating that the entire formula represents the coordination compound. Arrows point to different parts of the formula, providing additional information. The label "Central metal ion" points to Cu, indicating that copper is the central metal ion. "Ligand coordination number (1 + 5 = 6)" points towards $(\text{NH}_3)_5$ and Cl within the brackets, specifying that there are six ligands coordinated to the central metal ion. Lastly, the label "Free anions" points to Cl_2 outside the bracket, indicating that these are free anions. This visual breaks down the components of a complex ion, highlighting the relationships and elements involved in such structures.

[Generated by AI]

Try the worked examples and the exercise question on charges on complex ions.

Worked example 1

Cobalt(III) can form a complex ion with four ammonia and two chloride ligands. Calculate the overall charge and formula of the complex ion.

Charge of cobalt(III) is 3+

Charge of ammonia as a neutral ligand is 0

Charge of chloride is 1-

The overall charge is $(+3) + 4(0) + 2(-1) = 1+$

The formula of the complex ion is $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$

Worked example 2

Deduce the charge of chromium in $[\text{CrCl}(\text{H}_2\text{O})_5]\text{Cl}_2 \cdot \text{H}_2\text{O}$.

The overall charge is neutral, so:

$$\begin{aligned}(\text{Cr}) + 6(\text{H}_2\text{O}) + 3(\text{Cl}) &= 0 \\(\text{Cr}) + 6(0) + 3(-1) &= 0 \\ \text{Cr} &= 3+\end{aligned}$$



Exercise 1

Click a question to answer



1. A complex ion of $[\text{CrCl}_2(\text{NH}_3)_4]^+$ has chromium as a central metal ion with the charge of _____, and chlorine as well as ammonia as ligands.

☒ Check answers

Higher level (HL)

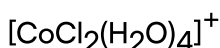
Activity

- IB learner profile attribute: Communicator
- Approaches to learning:

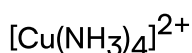
- Communication skills — Using digital media for communicating information
- Research skills — Evaluating information sources for accuracy, bias, credibility and relevance
- Time required to complete activity: 30—40 minutes
- Activity type: Group activity

Your task

Form a group of three. Each person researches one of the three complex ions:



or



or



1. After you have selected one of these three complex ions, research the application of your complex ion in industry.
2. Next, create a digital animation showcasing the structure of your chosen complex ion. This animation should accurately display the molecular geometry and the location of the lone pairs. There are various websites available you can use to create your videos or digital animations (e.g. [Flipgrid](https://info.flip.com/)  (<https://info.flip.com/>), [Animaker](https://www.animaker.com/)  (<https://www.animaker.com/>)).

Work collaboratively with your group to check that the information you give is accurate, such as the number of lone pairs of the ligands and the structure of the complex ion. Discuss any further suggestions and check for false information in your explanation to ensure everyone has a chance to participate in the evaluation process.